

My title*

My subtitle if needed

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First sentence. Second sentence. Third sentence. Fourth sentence.

1 Introduction

Overview paragraph

Estimand paragraph

Results paragraph

Why it matters paragraph

Telegraphing paragraph: The remainder of this paper is structured as follows. Section [2](#)....

2 Data

2.1 Overview

We use the statistical programming language R (R Core Team 2026).... Our data (Toronto Public Health 2026).... Following Alexander (2023), we consider...

Overview text

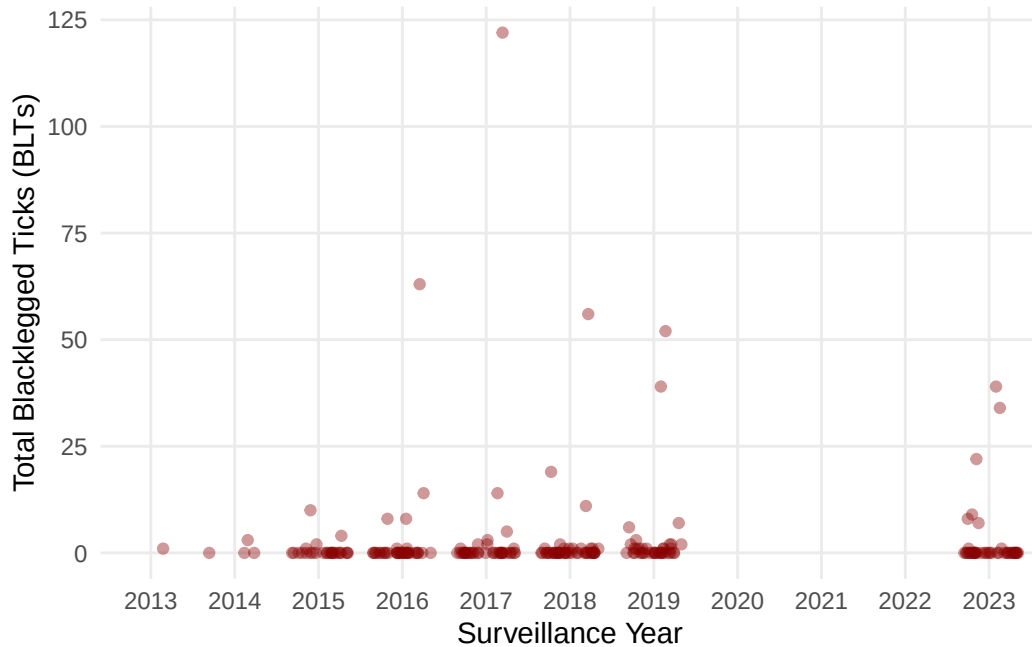


Figure 1: Distribution of total blacklegged ticks collected per park across surveillance years in Toronto.

2.2 Measurement

Some paragraphs about how we go from a phenomena in the world to an entry in the dataset.

```
# Aggregate ticks per park, filter for top 10, and sort ascending for horizontal display
top10_parks <- analysis_data |>
  group_by(park_locations) |>
  summarize(
    total_ticks = sum(total_blts, na.rm = TRUE),
    .groups = "drop"
  ) |>
  filter(total_ticks >= 1) |>
  slice_max(order_by = total_ticks, n = 10) |>
  mutate(
    park_locations = fct_reorder(park_locations, total_ticks)
  )

# Horizontal bar chart
```

*Code and data are available at: https://github.com/UpCatch/Toronto_Tick_Surveillance_Data.

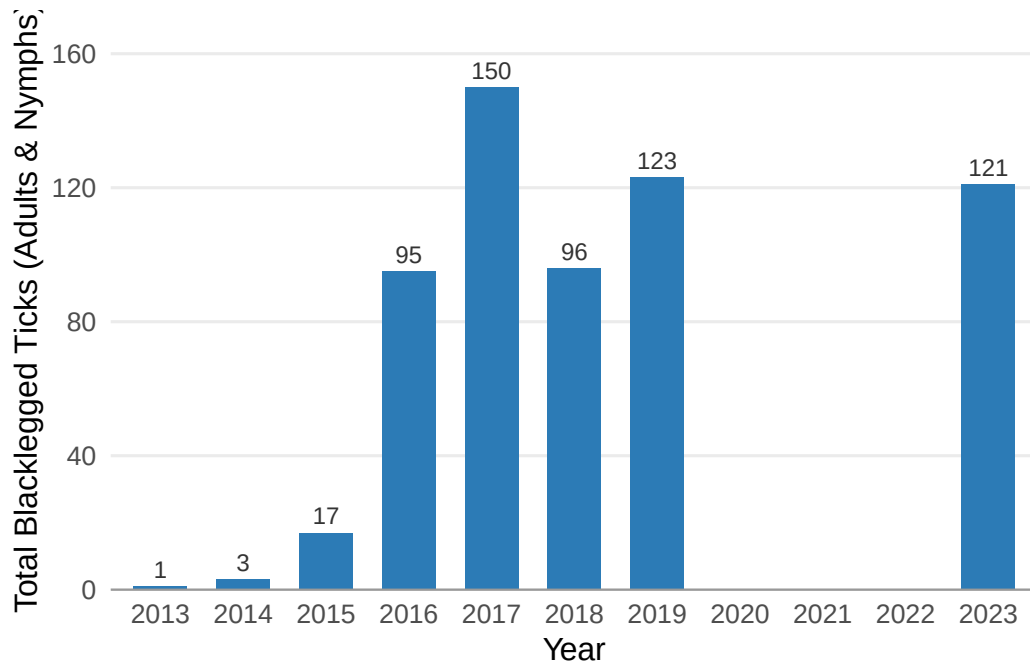


Figure 2: Total adult and nymph blacklegged ticks collected per year across all surveyed parks in Toronto.

```
ggplot(top10_parks, aes(x = total_ticks, y = park_locations)) +
  geom_col(
    fill = "#8b0000",
    width = 0.65
  ) +
  geom_text(
    aes(label = scales::comma(total_ticks)),
    hjust = -0.2,
    size = 3.5,
    color = "grey20"
  ) +
  scale_x_continuous(
    expand = expansion(mult = c(0, 0.12)),
    labels = scales::comma
  ) +
  labs(
    x = "Total Blacklegged Ticks (Adults & Nymphs)",
    y = NULL,
    title = "Top 10 Toronto Parks by Blacklegged Tick Counts",
  )
```

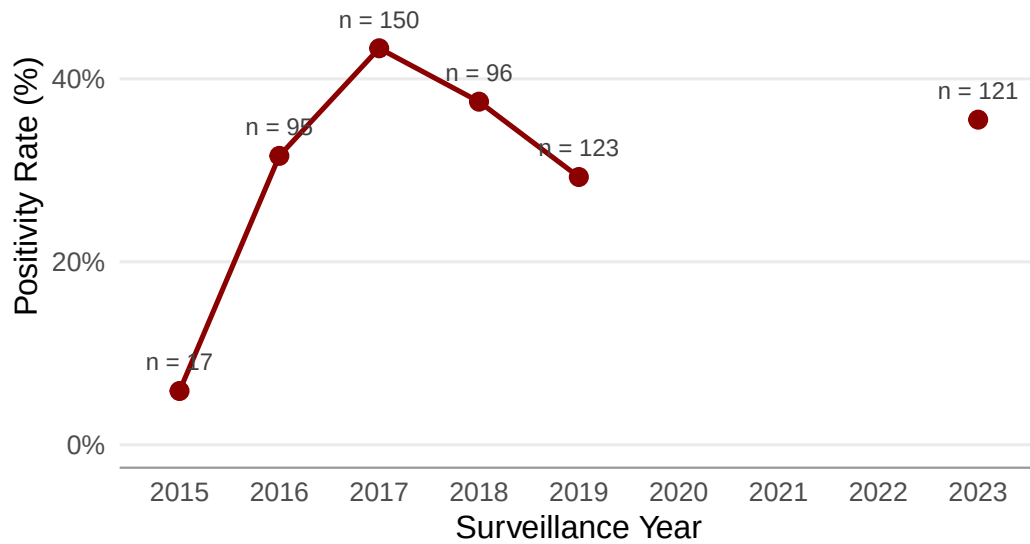
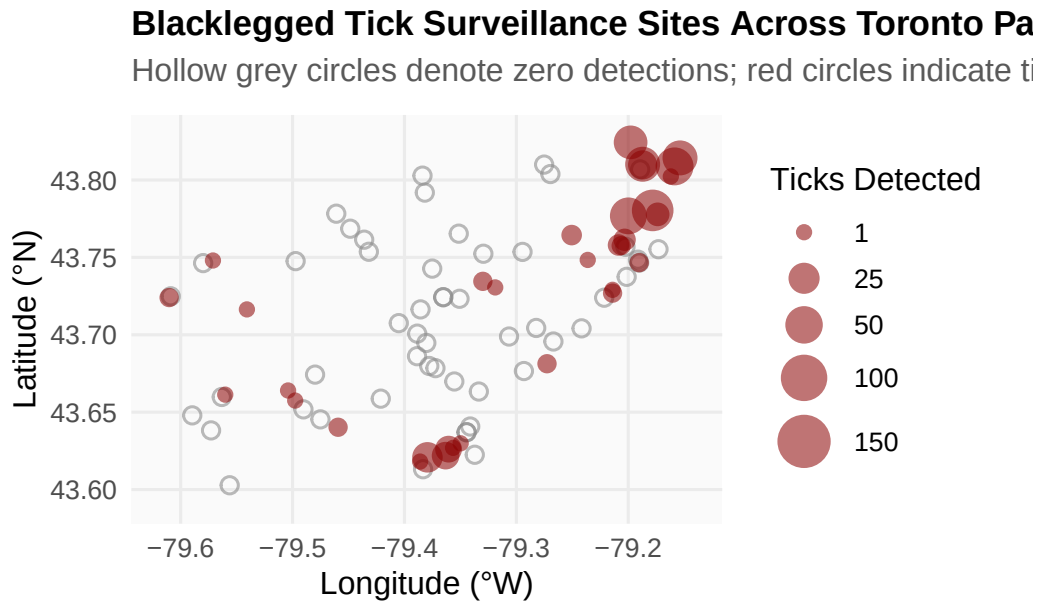


Figure 3: Annual blacklegged tick positivity rate across Toronto parks (2015–2023), with the number of positive ticks annotated.

```

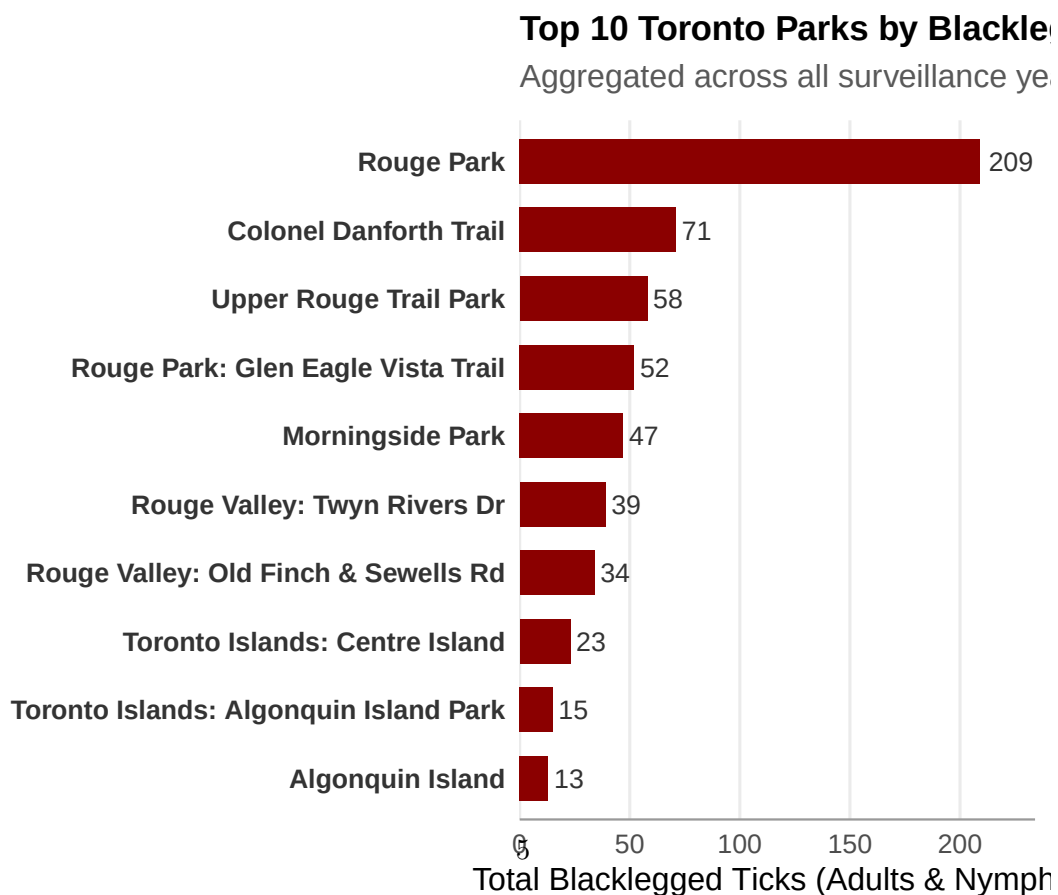
    subtitle = "Aggregated across all surveillance years (2013-2019, 2023)",
    caption = "Data: Toronto Public Health Blacklegged Tick Surveillance"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    panel.grid.minor = element_blank(),
    panel.grid.major.y = element_blank(),
    axis.line.x = element_line(color = "grey60", linewidth = 0.4),
    axis.text.y = element_text(size = 10, face = "bold", color = "grey20"),
    plot.title = element_text(face = "bold", size = 13),
    plot.subtitle = element_text(color = "grey35", margin = margin(b = 10)),
    plot.caption = element_text(color = "grey40", size = 8, margin = margin(t = 10))
  )

```



Note: Each dot is a summary over all years.

Figure 4: Geographic distribution of blacklegged tick surveillance sites in Toronto. Each dot represents a park summary aggregated over all surveillance years.



Data: Toronto Public Health Blacklegged Tick Surveillance

Figure 5: Top 10 Toronto parks with the highest cumulative blacklegged tick counts across all surveillance years.

References

- Alexander, Rohan. 2023. *Telling Stories with Data*. Chapman; Hall/CRC. <https://tellingstorieswithdata.com/>.
- R Core Team. 2026. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. <https://doi.org/10.32614/R.manuals>.
- Toronto Public Health. 2026. *Blacklegged Tick Surveillance*. <https://open.toronto.ca/dataset/blacklegged-tick-surveillance/>.